

# An autonomous career system for embodied creative research and development

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Career development for a specific research-and-development role is usually managed as motivation, memory, and scattered artifacts rather than as an auditable control problem. We report an autonomous career system that converts the Walt Disney Imagineering Research and Development mechanical Imagineering target into a logged state machine with source registries, role-fit dimensions, guardrails, review cycles, current-step selection, private current-work signals, and continuous manuscript updates. The live reviewer can critique public AO Labs evidence, the WDI role context, Disney Research context, CV material, project surfaces, and selected private source-status summaries, but it cannot send outreach, claim relationships, submit applications, or expose private source text without approval. At the 31 August 2026 evidence lock, the previously tracked Disney Careers lead returned HTTP 404 on the clicked page. The current interface does not claim a completed prototype or a current Disney lead. It keeps Imagineer as a live read surface while Brain, Progress, and PhD provide current-work, public-ledger, and note/file state. The main page shows the current action and the lead-verification control without mirroring raw Brain or PhD text. The refresh did not add a new AI review, application, outreach event, interview, referral, Disney relationship, external validation, or completed prototype. The live fit signal is 60 of 100 with early-but-directionally-relevant confidence, and the weakest signal is principal-scope ownership at 34 of 100. The contribution is not a job-success claim. It is a falsifiable protocol for making an ambiguous career-conversion process measurable, source-bounded, adaptive, and continuously documented while plainly exposing weak progress.

## 1 Introduction

Tool-using language-model systems can now reason, call external services, write artifacts, and update deployed software (1–4). These capabilities create an opportunity to treat long-horizon personal objectives as monitored systems rather than as episodic intentions, in the older spirit of autonomic systems and closed-loop discovery workflows (5, 6). A career target is especially difficult because the desired outcome depends on evidence quality, technical fit, social trust, timing, applications, and human judgment. Without a state model, repeated effort can feel intense while remaining scientifically uninterpretable.

This paper describes an early deployment of an autonomous career-signal system aimed at WDI Research and Development mechanical Imagineering roles. The system now treats the Principal R&D Imagineer - Mechanical Engineer profile as the north-star target. On 31 August 2026, the previously tracked Disney Careers destination for the Principal Ride Development Engineer posting, Disney job ID 10134485, returned HTTP 404 on the clicked page, so the system no longer treats it as a current lead (7). The active lead state is the exact search target `site:disneycareers.com Walt Disney Imagineering mechanical R&D Glendale`; this is not an application, referral, relationship, hiring claim, or permission for outreach. The positioning line is deliberately stable: mechanical PhD, soft robotics, creative prototyping, and AI-assisted tools for physical interaction systems.

The design borrows from controlled revenue experimentation: define a measurable state, choose the current bottleneck, constrain allowed actions, record interventions, and update a paper from real source material rather than from aspiration. The system is not allowed to invent credentials, inflate relationships, send spam, or apply on behalf of the operator. It can monitor state, update dashboards, maintain a paper, identify the next constraint, run a source-bounded AI review, and create or improve public artifacts that are grounded in existing work. The paper is not a one-time publication draft. It is a cumulative methods record, and it must change whenever the system changes in a way that affects evidence, policy, interface, metrics, sources, guardrails, or interpretation.

## 2 System architecture

The deployment consists of a public dashboard, a Railway-hosted backend, a JSON runtime state file mounted on a persistent volume, Brain and PhD source reads, Progress work-event and summary reads, and scheduled or operator-triggered automation. The backend exposes health, operations-check,

journal, weekly-paper, event, source-read, lead-check, daily-cycle, and AI-review behavior. As 54  
of 24 July 2026, the dashboard home page is intentionally a single action surface rather than an 55  
analysis surface. It ranks only recent career-relevant research signals, uses Brain for the most specific 56  
current-work note, uses Progress events and saved paper state for corroboration, and never promotes an 57  
older project merely because that project still appears in the event window. The profile page carries the 58  
longer state analysis and reads live backend, lead-check, and Progress timestamps instead of depending 59  
on a static profile date. Progress carries the source inventory, scan history, papers, private-source 60  
status, YouTube source, Curtis practice state, Relay state, CV, and sibling-app health. 61

The reviewer is a separate loop rather than a human-contact substitute. It builds a source bundle from 62  
the live state, the WDI mechanical R&D search target, the expired WDI R&D mechanical role-shape 63  
reference, the WDI R&D profile, the paper PDF, the CV, Sarrus, FluxCell, Relay, Relay Live, Progress, 64  
Curtis, Ocean, Talk, Nerve, Duet, Violin, Yum, Lily, AO Labs home, Disney Research roster, and a 65  
Disney Research bipedal-character paper page. It also includes a compact identity profile summarizing 66  
the operator as a mechanical engineering PhD candidate building soft robotic materials, reconfigurable 67  
mechanisms, motion prototypes, human-facing physical systems, public project surfaces, research 68  
papers, and autonomous systems. The model returns source-bounded JSON: verdict, score, current 69  
constraint, signals, open limitations, candidate system work, and approval boundaries. If the model 70  
fails or times out, a deterministic fallback reviewer produces a conservative critique. The stored record 71  
can keep the internal fallback cause, but public compact review payloads expose only a sanitized 72  
fallback state and a plain fallback note rather than raw provider text. 73

**Table 1.** State variables at the current evidence lock. Values are reported from the live operations endpoint, the current 2026-W36 weekly-paper preview, Brain current-work metadata, the Progress summary, and the clicked Disney Careers destination check on 31 August 2026.

| Variable                    | Recorded value                                                                                                                                                                                                                         |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target company and location | Walt Disney Imagineering, Glendale, California; R&D remains the north-star lane                                                                                                                                                        |
| Live rung                   | No current Disney lead verified; WDI R&D principal title remains north-star only                                                                                                                                                       |
| North star                  | Principal R&D Imagineer - Mechanical Engineer                                                                                                                                                                                          |
| Fit signal                  | 60 of 100 with early-but-directionally-relevant confidence                                                                                                                                                                             |
| Weekly snapshot             | 2026-W30; status <code>live_preview_until_paper_snapshot</code> ; the live weekly endpoint generates the current preview, and the public PDF route carries the deployed manuscript snapshot for this evidence lock                     |
| Dominant bottleneck         | Principal signal: visible ownership, technical direction, source depth, and approval-gated external validation                                                                                                                         |
| Counters                    | Seven work-event logs; zero outreach events; one daily cycle; fifteen AI-review records; four portfolio anchors; five profile-prepared artifacts; thirty-three journal entries in the live operations state at the current source read |
| Active experiment           | Autonomous career loop v0                                                                                                                                                                                                              |
| Reviewer source graph       | Operations endpoint exposes the profile and reviewer source graph; this lead-expiry refresh did not create a fresh AI review                                                                                                           |
| Human approval boundary     | Applications, sensitive outreach, referrals, claims involving real people, and any private contact/email data                                                                                                                          |
| Progress source registry    | Progress remains the current source-state input for the profile; private source text remains summarized by status and freshness rather than mirrored publicly                                                                          |
| Profile freshness           | Public profile exposes the live read timestamp, source count, fit score, bottleneck score, current system action, and six role-fit lanes; the Imagineer home page shows only the current step                                          |
| Paper continuity rule       | Any substantive change to a paper-backed system updates the manuscript and PDF in the same work cycle                                                                                                                                  |

### 3 Decision policy

The policy now separates state scoring from next-step selection. State scoring still reports six role-fit dimensions: mechanical depth, creative prototyping, human-facing motion, principal signal, Glendale application profile, and the autonomous system itself. Each dimension reports the tracked signal, basis, and source-backed limitation that would move the score. The score basis is intentionally calibrated: a base prior from the state file, a bounded event bonus, a bounded portfolio bonus, a daily-cycle bonus, optional review-loop credit, and a readiness ceiling that names the unresolved blocker. A score of 100

is reserved for no known blocker in that lane, not for accumulated activity.

The next-step selector uses a failure-cost evidence-ROI policy. For each candidate action, the system assigns points for direct Disney-role alignment, relief of the current bottleneck, creation of current inspectable evidence, compounding value for future reviews, and urgency. It subtracts friction and approval-gate penalty. The selected action is the highest-scoring move that can start immediately without external outreach. This change matters because the weakest-signal rule alone can choose generic analysis work. The new policy favors actions that change the public record or the experiment state, especially when the current failure point is principal-scope ownership.

In the current lock, the weakest live signal is principal signal, but the selected action must follow the active research horizon rather than a distant endpoint. The previous selector violated that rule by scanning backward until it found any WaveVis-shaped event and by retaining a second hard-coded WaveVis fallback, even when Brain and Progress contained newer Cellular Soft Robots work. The corrected selector ranks recent career-relevant research lanes by authoritative source time, uses the most specific Brain note as the action basis, and uses Progress work events and saved paper state as corroboration. At this evidence lock, the selected action is to lock the Cellular Soft Robots pressure-test protocol: define and document one zero-psi conditioning rule, then run the next pressure-width trials under that same protocol before using the data in the paper. This is an experiment-quality and manuscript-trust gate, not a claim that the experiment has already been completed or externally validated.

This policy is intentionally conservative. It can make public artifacts, update the dashboard, refresh the manuscript, run autonomous critique, and record state. It cannot manufacture the social trust that a principal-level role requires. If an action requires a real person's attention or reputation, the system must stop at preparation and request approval before sending. The review loop therefore becomes a repeatable calibration instrument, not a license to automate human outreach.

## 4 Results

The current deployment is functional and has moved beyond the initial dashboard-only state. The dashboard and backend are live, the weekly-paper endpoint exposes the live preview, the site exposes a PDF manuscript, and the AI-review path has produced live production critiques from the public source graph. The current update reports 60 of 100 fit, early-but-directionally-relevant confidence,

principal signal as the current bottleneck at 34 of 100, and zero outreach events. The current refresh did not create a new model review, application, interview, referral, contact, or external validation. In the latest recorded production review cycle, the then-configured GPT-5.5 call returned an OpenAI quota error, so the deterministic fallback produced the conservative review: score 74 of 100, credible mechanism depth, developing Disney motion signal, and thin principal-scope ownership signal. The runtime default has since been lowered to GPT-5 mini for cost control, while an environment override can still select a stronger model for a deliberate review.

The largest artifact change in this cycle was the Sarrus technical record. The system published geometry, compliant-link and flexural-joint path, pressure-driven expansion, module assembly, build state, force/stiffness figures, and motion examples. It then added an HTML-visible measurement table with pressure range, expansion ratio, estimated leg angle, output force, local stiffness, hysteresis status, planar dome height, double-dome resolution, and prototype limitation. The WDI R&D profile now uses this table as the dominant technical path while preserving the current data boundary: raw digitized force, stiffness, and hysteresis curves with uncertainty remain unresolved.

After the latest interface revision, the live site uses a compact operating surface instead of explanatory tiles or second-person coaching. The state surface now reports compact facts: Mechanical depth, principal signal, Progress-backed source state, profile freshness, and the boundary around applications, referrals, direct outreach, and person-facing claims. The type scale was reduced because the dashboard is an operating surface, not a poster. The outcome-signal cards use the same standard: direct signal, current source state, and system-owned operation. This is a functional requirement, not a tone preference.

The system-level fit signal is now calibrated to 60 of 100 after the tracked Disney Careers destination stopped verifying. It is an internal readiness signal, not an external hiring probability. The review result is deliberately separate: it scores the current public profile, not the probability of being hired. It confirms that the overall direction is credible for WDI mechanical design while rejecting the idea that the current public profile is already principal-level. The profile now includes both a principal-scope boundary and an above-fold role-verification banner: no active Glendale WDI ride-development lead is verified, while the WDI R&D principal title remains a north-star profile until principal-scope leadership work is independently verified.

The CV was revised in the same paper cycle because it is part of the reviewer source graph and the public application record. The selected-public-work section now treats Sarrus as one coherent

project rather than splitting it into a separate subpage link. The top line and profile now point toward mechanical R&D, soft robotics, physical interaction, motion systems, and human-facing physical prototypes without naming the target directly. The Sarrus research and publication entries now use Sarrus-linkage soft-robotic cells and planar/cylindrical/locomotion assemblies instead of a generic topology frame. This keeps the CV aligned with the WDI R&D objective without reading like a named-company pitch or overstating principal-scope evidence.

The progress ledger adds a cross-application memory layer rather than replacing the Imagineer dashboard. Its first deployed version scanned eighteen public sources; the 24 July 2026 Progress summary reports 92 tracked public sources and 88 healthy sources, while the Imagineer operations profile lists 33 profile sources and the reviewer source list reports 31 sources. The source graph includes project pages, papers, the CV surface, Progress state, Brain health, private-source status summaries, WaveVis routes, Blood, Paper, Cooking, Yum, Violin, and sibling AO Labs surfaces, while private working text is exposed only as status, size, and freshness on public routes. Progress returns the Imagineer current step as structured JSON, but it no longer searches backward for a WaveVis event before considering the newest research lane. This creates the storage surface needed for long-term trend comparisons across papers, practice, portfolio state, public pages, planning logs, and outcome systems without publishing the private working log.

The Imagineer main page was reduced in the same cycle. The previous home page attempted to show paper state, WDI state, lane scores, loop health, and signal analysis at once. That analysis now belongs on the profile page and in Progress. The home page now has one purpose: show the current small step, the blunt reason it matters, the expected time cost, the freshness timestamp, and a direct path to the active work surface. The current step now combines Brain current-work notes, the ten newest Progress research events, and Progress saved-source timestamps; only career-relevant research lanes within a 30-day horizon are eligible, and unrelated newer app events are ignored. A generic loading state replaces the old WaveVis fallback, so a temporary source timeout cannot present a cached project as current. The client allows 45 seconds for this live composition, while the narrowed Progress event read keeps the current event available under the real ledger load. PhD remains a note and file capture surface, and the only source-maintenance control on the home surface is clicked-destination lead verification. The API reports source counts and freshness, derives public-safe topic flags, and keeps raw Brain and PhD note text out of the public Imagineer surface. This readout treats the current Cellular Soft Robots experiment and paper gate as a controllable R&D lever while preserving the



boundary between active research progress and completed external validation.

The profile page now exposes a live read timestamp rather than the older static 9 May review timestamp. It reads the Imagineer operations endpoint and the Progress source ledger, then renders source count, readiness score, bottleneck score, current system action, source freshness, system metrics, and the six role-fit lanes. This matters because the profile is not only a Disney-specific pitch; it is a compressed read of the public AO Labs record. New systems such as Curtis, unrelated project surfaces such as Yum, updates to Sarrus, paper endpoints, and progress-ledger snapshots are counted as context about the operator's range, persistence, taste, and systems-building pattern. They do not automatically increase role-fit scores. The scoring boundary remains explicit: only source material that strengthens mechanical R&D, physical interaction, prototype execution, systems ownership, or principal-scope responsibility moves the Imagineer readiness state.

**Table 2.** Latest autonomous review result. The reviewer path is source-bounded over the public AO Labs graph; the latest production record used the deterministic fallback because the then-configured GPT-5.5 call returned an OpenAI quota error. The runtime default is now GPT-5 mini unless overridden.

| Field               | Recorded value                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model               | Deterministic fallback after OpenAI quota error; runtime default now GPT-5 mini unless overridden                                                                                                                                                                                                                                                                                    |
| Scope               | Whole public AO Labs graph plus WDI role and Disney Research context                                                                                                                                                                                                                                                                                                                 |
| Source count        | Progress-backed source state plus the Imagineer profile and reviewer source graph; no fresh AI review was run in the 31 August lead-expiry refresh                                                                                                                                                                                                                                   |
| Review score        | 60 of 100                                                                                                                                                                                                                                                                                                                                                                            |
| Verdict             | Credible core, not inevitable yet                                                                                                                                                                                                                                                                                                                                                    |
| Top issue           | Sarrus makes the mechanical case credible; principal-level ownership is not visible enough yet                                                                                                                                                                                                                                                                                       |
| Current system work | Single-step home page with a Brain-plus-Progress current action, career-source, income-path, and A3 car-path readout; private Brain/PhD text suppressed from public output; sanitized public reviewer state; clicked-destination lead verification; profile analysis surface; Progress source ledger; YouTube/Curtis source tracking; AO Labs Progress tile; continuous paper update |



**Table 3.** Role-fit dimensions after the current autonomous review cycle. Scores are internal operating metrics, not external hiring probabilities; the live dashboard exposes the tracked signal, basis, and unresolved signal for each row.

| Dimension            | Score | Current interpretation                                                                                                                                          |
|----------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mechanical depth     | 86    | Strong but not complete; Sarrus has a public technical record and measurement table, with raw digitized curves, uncertainty, and test conditions still pending. |
| Creative prototyping | 82    | Strongest signal; attached to concrete physical demonstrations.                                                                                                 |
| Human-facing motion  | 79    | Credible signal, but still needs a concise guest-facing motion sequence.                                                                                        |
| Principal signal     | 34    | Current bottleneck; visible ownership, technical direction, collaborators, and external validation remain thin.                                                 |
| Glendale profile     | 72    | Credible for the active rung; CV framing is now aligned, while demo reel and role-specific narrative remain open.                                               |
| Autonomous system    | 72    | Backend, dashboard, durable state, reviewer, journal, and paper loop are live, but scheduled runs, evaluation history, and outcome tracking are early.          |

5 Limitations and guardrails

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The system cannot guarantee an offer. A job decision depends on timing, hiring needs, competition, interview performance, portfolio interpretation, and institutional context. The useful claim is narrower: the system can make progress legible, choose the next bottleneck, prevent fake progress, and turn repeated effort into logged artifacts. Evaluation depends on public artifact velocity, critique quality, application readiness, interviews, and eventual conversion.

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The guardrails are part of the result. The system forbids fabricated credentials, fake outreach, invented Disney relationships, automated applications, and claims that are not supported by public or logged sources. Private email/contact data is excluded from the reviewer unless explicitly approved. These constraints reduce short-term aggressiveness, but they preserve interpretability. Strong progress must be traceable to artifacts, logs, AI reviews, human approvals, applications, or outcomes.

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6 Availability

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The live dashboard is available at <https://imagineer.aolabs.io/>, with a fallback at <https://aolabs.io/imagineer/>. The live backend uses the base route <https://imagineer.aolabs.io/api/imagineer/> and exposes ops-check, weekly-paper, ai-review, events, and lead-check/run; PhD source reads are surfaced through ops-check. This manuscript is a public artifact and must be refreshed from logged sources in the same work cycle as substantive changes to the dashboard, backend, reviewer policy, source graph, public profile, Sarrus/portfolio evidence, metrics, or guardrails.

## References

1. S. Yao *et al.*, ReAct: Synergizing reasoning and acting in language models. *International Conference on Learning Representations* (2023).
2. T. Schick *et al.*, Toolformer: Language models can teach themselves to use tools. *arXiv preprint arXiv:2302.04761* (2023).
3. Q. Wu *et al.*, AutoGen: Enabling next-gen LLM applications via multi-agent conversation framework. *arXiv preprint arXiv:2308.08155* (2023).
4. L. Wang *et al.*, A survey on large language model based autonomous agents. *arXiv preprint arXiv:2308.11432* (2023).
5. J. O. Kephart, D. M. Chess, The vision of autonomic computing. *Computer* **36**, 41–50 (2003).
6. A. G. Kusne *et al.*, On-the-fly closed-loop materials discovery via Bayesian active learning. *Nature Communications* **11**, 5966 (2020).
7. The Walt Disney Company, *Principal Ride Development Engineer*, Clicked destination returned HTTP 404 on 31 August 2026. Use the exact Disney Careers search target instead; this is not an application, referral, relationship, or hiring claim. (2026; <https://www.disneycareers.com/en/job/glendale/principal-ride-development-engineer/391/87268384416>).